MCQ A · one-sided limits and transformations

After S24; MCQ B after S26.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. $\lim_{x\to 2^-} \frac{1}{x-2}$ is:

A. $+\infty$ B. $-\infty$

C. 0 D. 2

2. For $x < 0$, $\sqrt{x^2}$ is:

A. x B. x^2

C. $-x$ D. $\pm x$

3. $\lim_{x\to+\infty} \frac{3x^2-x}{x^2+4}$ is:

A. 0 B. $+\infty$

C. -1 D. 3

4. $\lim_{x\to 1} \frac{x^2-1}{x-1}$ is:

A. 2 B. 0

C. $+\infty$ D. does not exist

5. For $f(x) = 1 + \frac{\sin x}{x}$, $x > 0$, the line $y = 1$:

A. is not an asymptote B. is an asymptote with no intersection

C. is an asymptote crossed infinitely often D. is tangent at every point

MCQ B · comparative growth and choice of tool

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. $\lim_{x\to+\infty} x^2 e^{-x}$ is:

A. $+\infty$ B. 0

C. 1 D. $-\infty$

7. $\lim_{x\to-\infty} xe^x$ is:

A. 0 B. $+\infty$

C. $-\infty$ D. 1

8. The appropriate tool for $\sqrt{x^2+2x} - x$ at $+\infty$ is:

A. the binomial B. the IVT

C. differentiation D. the conjugate

9. If $f(x) \to 0$ and $g(x) \to +\infty$the product:

A. always tends to 0 B. always tends to $+\infty$

C. needs further study D. is impossible

10. If $f(x) \geq x^2$ for $x \geq 1$, then:

A. $f(x)/x^2 \to 1$ B. $f(x) \to +\infty$

C. f is increasing D. $f(x) - x^2 \to 0$

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. A graph never crosses its horizontal asymptote.

2. A finite limit at a requires that $f(a)$ is defined.

3. The symbol ∞/∞ gives a limit equal to 1.